Command Line Utilities-I

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Linux Command Line Utilities

Why do we need to study them?

- Using CLI is faster than using a GUI. e.g.:
- Remote access. e.g.: Server Management
- Scripting. e.g.: Installing packages across multiple devices

and many more...

Linux provides a range of tools for managing and configuring networks.
These include a range of GUIs and powerful command-line tools such as **netstat** and **ifconfig**.

Let’s cover few of them here….

How to know the details about the network interfaces of your device using the command line?

ifconfig

- Used to configure network interfaces
- It is **usually** used for debugging or when system tuning is needed
- If no arguments are given, ifconfig displays the status of the currently active interfaces

```
ryzen02@ryzen02-rs-iiitd:~$ ifconfig
cni0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1450
        inet 10.244.0.1  netmask 255.255.255.0  broadcast 10.244.0.255
        inet6 fe80::34d5:88ff:fee9:3bec  prefixlen 64  scopeid 0x20<link>
        ether 36:d5:88:e9:3b:ec  txqueuelen 1000  (Ethernet)
        RX packets 1913786467  bytes 2436890814504 (2.4 TB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1880755822  bytes 2359501999374 (2.3 TB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether 02:42:ea:d7:09:a9  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp5s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.28.126  netmask 255.255.240.0  broadcast 192.168.31.255
        inet6 fe80::d5c8:8e7c:6397:a6be  prefixlen 64  scopeid 0x20<link>
        ether 7c:10:c9:d1:11:b8  txqueuelen 1000  (Ethernet)
        RX packets 131048793  bytes 40191532694 (40.1 GB)
        RX errors 0  dropped 0  overruns 75  frame 0
        TX packets 11496990  bytes 4428963994 (4.4 GB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device memory 0xed700000-ed71ffff

flannel.1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1450
        inet 10.244.0.0  netmask 255.255.255.255  broadcast 0.0.0.0
        inet6 fe80::18bf:e1ff:fec9:4fdb  prefixlen 64  scopeid 0x20<link>
        ether 1a:bf:e1:c9:4f:db  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 3496 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 300799655  bytes 51506736322 (51.5 GB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 300799655  bytes 51506736322 (51.5 GB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

- To view all the interfaces (it will display the inactives ones too)add the flag -**a**
  e.g. **ifconfig -a**
- To activate an interface: **ifconfig <if-name> up**
- To deactivate an interface: **ifconfig <if-name> down**
- Few other configurations you can perform on the network interfaces

IP	**Ifconfig <if-name> <new-ip-addr>**
Netmask	**Ifconfig <if-name> broadcast <new-broadcast>**
Broadcast	**Ifconfig <if-name> netmask <new-netmask>**
mtu	**Ifconfig <if-name> mtu <new-mtu>**

For complete details on the command ifconfig run the following command: **man ifconfig**

How to find the round trip time between your computer and a remote device using the command?

ping

- Ping is short for **Packet Internet Groper**, mainly used for checking the network connectivity among host/server and host.
- Ping works by sending one or more **ICMP** (Internet Control Message Protocol) **Echo Request** packets to a specified destination IP on the network and waits for reply. When the destination receives the packet, it responds with an ICMP echo reply.

```
ryzen02@ryzen02-rs-iiitd:~$ ping google.com
PING google.com (142.250.206.174) 56(84) bytes of data.
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=1 ttl=118 time=3.88 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=2 ttl=118 time=3.87 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=3 ttl=118 time=3.77 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=4 ttl=118 time=3.77 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=5 ttl=118 time=3.76 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=6 ttl=118 time=3.80 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=7 ttl=118 time=3.85 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=8 ttl=118 time=3.83 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=9 ttl=118 time=3.79 ms
64 bytes from del11s22-in-f14.1e100.net (142.250.206.174): icmp_seq=10 ttl=118 time=3.75 ms
^Z
[4]+  Stopped                 ping google.com
ryzen02@ryzen02-rs-iiitd:~$ clear
```

Functionality	Command
Host reachable or not	ping google.com
To specify the number of Echo Request packages to be sent	ping -c 5 google.com
To induce a time delay between every packet being send	ping -i 5 google.com
To specify number of data bytes to be sent	ping -s 1000 google.com
To send pkts from a particular interface	ping -I eth0 google.com

For more details on ping: **man ping**

Is it possible to investigate the path that network packets take?

traceroute

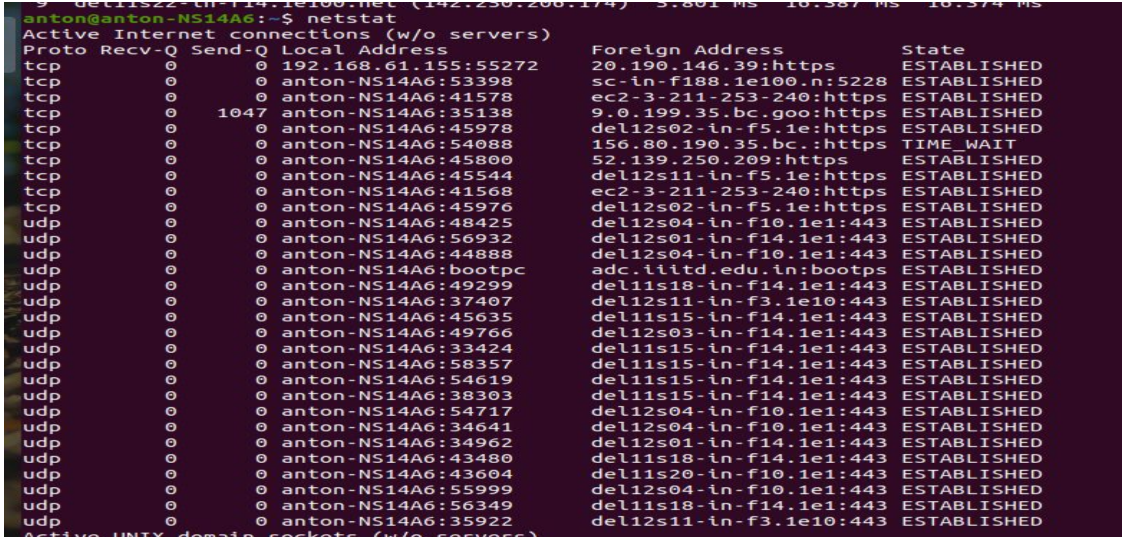
- Traceroute is a command-line utility for tracing the full path from your local system to another network system.
- Prints the number of hops (router IPs) in that path you travel to reach the end server.
- By default it sends an order of UDP packets, routes three packets of data to test each hop

```
[ryzen02@ryzen02-rs-iiitd:~$ traceroute google.com
traceroute to google.com (142.250.206.174), 64 hops max
  1   192.168.16.254  0.662ms  0.736ms  0.793ms
  2   192.168.1.99  0.080ms  0.071ms  0.093ms
  3   103.25.231.1  0.297ms  0.193ms  0.204ms
  4   *  *  *
  5   10.119.234.162  2.963ms  2.723ms  2.809ms
  6   72.14.195.56  5.131ms  6.097ms  6.508ms
  7   74.125.243.97  4.652ms  5.841ms  4.373ms
  8   142.251.76.201  3.786ms  3.665ms  3.635ms
  9   142.250.206.174  3.746ms  3.747ms  3.687ms
ryzen02@ryzen02-rs-iiitd:~$
```

- *** in the output means ->
    - Intermediate Routers may be Overload
    - organizations configure their routers not to respond to traceroute because they don’t want to reveal details of their internal network
- To specify the type(udp/icmp) of packets to be sent: **traceroute -- type=icmp google.com**
- To specify the maximum number of hops: **traceroute -m 10 google.com**
- To check the other flags that can be used: **traceroute --help**

netstat

- ***netstat*** : (short for network statistics) is a command line network utility.
- It is used to display information about network connections, routing tables, interface statistics and masquerade connections.
- It is available in Windows Linux and macOS.



```
anton@anton-NS14A6:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 192.168.61.155:55272    20.190.146.39:https     ESTABLISHED
tcp        0      0 anton-NS14A6:53398      sc-in-f188.1e100.n:5228 ESTABLISHED
tcp        0      0 anton-NS14A6:41578      ec2-3-211-253-240:https ESTABLISHED
tcp        0   1047 anton-NS14A6:35138      9.0.199.35.bc.goo:https ESTABLISHED
tcp        0      0 anton-NS14A6:45978      del12s02-in-f5.1e:https ESTABLISHED
tcp        0      0 anton-NS14A6:54088      156.80.190.35.bc.:https TIME_WAIT
tcp        0      0 anton-NS14A6:45800      52.139.250.209:https    ESTABLISHED
tcp        0      0 anton-NS14A6:45544      del12s11-in-f5.1e:https ESTABLISHED
tcp        0      0 anton-NS14A6:41568      ec2-3-211-253-240:https ESTABLISHED
tcp        0      0 anton-NS14A6:45976      del12s02-in-f5.1e:https ESTABLISHED
udp        0      0 anton-NS14A6:48425      del12s04-in-f10.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:56932      del12s01-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:44888      del12s04-in-f10.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:bootpc     adc.iiitd.edu.in:bootps ESTABLISHED
udp        0      0 anton-NS14A6:49299      del11s18-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:37407      del12s11-in-f3.1e10:443 ESTABLISHED
udp        0      0 anton-NS14A6:45635      del11s15-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:49766      del12s03-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:33424      del11s15-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:58357      del11s15-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:54619      del11s15-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:38303      del11s15-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:54717      del12s04-in-f10.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:34641      del12s04-in-f10.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:34962      del12s01-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:43480      del11s18-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:43604      del11s20-in-f10.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:55999      del12s04-in-f10.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:56349      del11s18-in-f14.1e1:443 ESTABLISHED
udp        0      0 anton-NS14A6:35922      del12s11-in-f3.1e10:443 ESTABLISHED
```

Common flags of netstat are:

- -a or --all: Display all connections, including listening and non-listening sockets.
- -t or --tcp: Display TCP connections.
- -u or --udp: Display UDP connections.
- -p or --program: Display the process ID and name that's using the socket.
- -r or --route: Display the routing table.
- -i : displays statistics for the network interfaces currently configured.
- -l or --listening: Display only listening sockets.

DNS

Before going to next two commands we need to get some idea about **DNS.**

- **DNS or Domain Name Server** is a system used to convert human-readable domain names (like https://iiitd.ac.in) into IP addresses (like 192.168.2.127) that computers understand.
- It acts like a "phone book" for the Internet, helping devices find each other using names instead of numbers.

Resolution Process:

1. A user enters a domain name (e.g., https://iiitd.ac.in) in a web browser.
2. The resolver sends a query to a DNS server to find the IP address associated with the domain name.
3. If the DNS server has the IP address in its database (cache), it responds immediately.
4. If not, the DNS server recursively queries other DNS servers until it finds the authoritative server for the domain.
5. The authoritative server provides the IP address, which is sent back to the user.

Types of DNS Records:

A Record (Address Record):

- Maps a domain name to an IPv4 address.
- Used to resolve domain names to their corresponding IP addresses.

AAAA Record (IPv6 Address Record):

- Maps a domain name to an IPv6 address.
- Similar to the A record but for IPv6 addresses.

MX Record (Mail Exchange Record):

- Specifies the mail servers responsible for receiving email messages for a domain.
- Used to route email traffic to the correct mail servers.

TXT Record (Text Record):

- Stores arbitrary text data associated with a domain.
- Can be used for various purposes, including domain verification, SPF records for email authentication, and more.

PTR Record (Pointer Record):

- Performs a reverse DNS lookup by mapping an IP address to a domain name.
- Used to verify the authenticity of the reverse mapping.

SOA Record (Start of Authority Record):

- Contains administrative information about the zone, including the primary authoritative name server and contact details and various timers.
- Refresh Interval: This specifies how often secondary DNS servers should check with the primary server for updates to the DNS zone data.
- Retry Interval: If a secondary DNS server fails to refresh the zone data during a refresh attempt, the retry interval specifies how long it should wait before trying again.
- Essential for zone transfer and maintaining consistency in DNS data.

nslookup

- ***nslookup*** stands for "Name Server Lookup."
- It is a command-line tool used to query the Domain Name System (DNS) to obtain information about domain names and IP addresses.
- With the nslookup command, you can perform various types of DNS queries, including:

  - Forward Lookup: Given a domain name, it returns the corresponding IP address.

    ```
    nslookup www.example.com
    ```

  - Reverse Lookup: Given an IP address, it returns the corresponding domain name (also known as a PTR record).

    ```
    nslookup 157.240.16.16
    ```

  - Querying Specific DNS Servers: You can specify the DNS server you want to query.

    ```
    nslookup www.meta.com 8.8.8.8/208.67.222.222
    ```

dig

- The dig command is a powerful DNS (Domain Name System) tool used in the Linux command-line environment to query DNS servers for information about domain names, IP addresses, and other DNS-related data.
- It is commonly used to troubleshoot DNS-related issues, gather DNS information, and perform various DNS-related tasks.
- Here are a few examples of how you might use the dig command:
  - Basic DNS Query:

    ```
    dig www.google.com
    ```

  - Query Specific Record Type:

    ```
    dig google.com MX
    ```

  - Query DNS Server Directly:

    ```
    dig @8.8.8.8 twitter.com
    ```

  - Reverse DNS Lookup (IP to Domain):

    ```
    dig -x 8.8.8.8
    ```

References

- nestat, Nslookup, dig from The Linux Documentation Project.
- Ping, traceroute, ifconfig